

Rank-6 fusion rings

A reproducible census through multiplicity 6

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Abstract

We explain the exact enumeration of all based fusion rings of rank 6 through multiplicity 6, comprising 5,799 isomorphism classes. The account separates the mathematical coverage argument, the completed search, and independent verification of the emitted tensors. All duality types are included, and no categorifiability filter is applied. Code, machine-readable counts, full multiplication tensors and provenance records accompany the text.

AI assistance and verification. GPT-6 Astra Pro assisted with code, exposition and computational checks. Exhaustive-search arguments and exact independent tensor verification are recorded separately; no proof-assistant certification or peer-reviewed acceptance is claimed.

Scope. A census of based rings, not a classification of tensor categories. Only completed whole-rank results are released.

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1 Objects, equivalence and the counting convention

A *fusion ring* here is a free abelian group with a distinguished finite basis $B = \{b_0, b_1, \dots, b_{r-1}\}$, an associative multiplication with unit b_0 , and an involution $i \mapsto i^*$ fixing 0. Its structure constants satisfy

$$b_i b_j = \sum_k N_{ij}^k b_k, \quad N_{ij}^k \in \mathbb{Z}_{\geq 0},$$

$$N_{0j}^k = N_{j0}^k = \delta_{jk}, \quad N_{ij}^0 = \delta_{i^*, j}, \quad (1)$$

$$N_{ij}^k = N_{i^*k}^j = N_{kj}^{i^*}. \quad (2)$$

These conventions agree with the usual based-ring definition; see [1, 2]. No assumption of categorifiability, commutativity or integrality of Frobenius–Perron dimensions is imposed unless proved in the relevant rank.

A based isomorphism is a bijection of the distinguished bases preserving the unit and multiplication. In tensor notation it is a permutation q fixing 0 such that

$$N_{ij}^k = N_{q(i)q(j)}^{q(k)} \quad \text{for every } i, j, k.$$

The coefficient of the unit determines the involution, so such an isomorphism automatically intertwines duality. The *multiplicity* is the maximum entry of the *full* ordered multiplication tensor. It is at least one, including for group rings. Put

$$c_r(m) = \#\{\text{rank-}r \text{ fusion rings of multiplicity exactly } m\} / \cong, \quad C_r(M) = \sum_{m=1}^M c_r(m).$$

A list through M contains all rings of multiplicity at most M , not only those of multiplicity equal to M . This distinction is essential when exporting OEIS b-files.

1.1 Why duality types can be searched separately

An involution on the $r - 1$ nonunit elements consists of p disjoint transpositions and $r - 1 - 2p$ fixed points, with $0 \leq p \leq \lfloor (r - 1)/2 \rfloor$. Fix one representative d_p of each type. An isomorphism between two tensors with involution d_p commutes with d_p . Hence it is sufficient to quotient that stratum by the centralizer

$$Z_{S_{r-1}}(d_p), \quad |Z_{S_{r-1}}(d_p)| = 2^p p! (r - 1 - 2p)!.$$

Different p cannot be isomorphic. Therefore a whole-rank census is complete exactly when every required stratum has been searched exhaustively. Finishing only the non-self-dual strata does not certify a whole-rank counting term.

2 The certified release at rank 6

The released bound is $M = 6$ and the cumulative number of classes is $C_6(6) = 5,799$. Every duality type is included. The accompanying full tensor file has one record per based-isomorphism class.

Dual pairs	Variables	Equations	Relabellings
0	35	55	120
1	30	75	12
2	25	85	8

Exact multiplicity m	$c_6(m)$
1	39
2	154
3	384
4	872
5	1,582
6	2,768

The complete-run evidence is available through `results/census.json` and the corresponding reports under `verification/`. Earlier ranks retain their original completed-run provenance; repository checks and newly executed runs are labelled separately.

3 An exhaustive finite-domain model

3.1 Coordinates directly from reciprocity

On nonunit triples apply the two transformations

$$(i, j, k) \mapsto (i^*, k, j), \quad (i, j, k) \mapsto (k, j^*, i).$$

Assign one variable $x_O \in \{0, \dots, M\}$ to each resulting orbit O . The unit entries and N_{ij}^0 are already fixed. Thus every tensor satisfying unit, duality and reciprocity occurs exactly once as an assignment of orbit coordinates. Three successive reciprocity transformations give $N_{ij}^k = N_{j^*i^*}^{k^*}$, so duality is an anti-involution. In particular, the self-dual stratum is commutative, at every rank.

For every nonunit i, j, k, ℓ impose the integer polynomial

$$\sum_s N_{ij}^s N_{sk}^\ell - \sum_s N_{jk}^s N_{is}^\ell = 0. \quad (3)$$

Associativity involving a unit is automatic. When $\ell = 0$, the two sides reduce to $N_{ij}^{k^*}$ and $N_{jk}^{i^*}$, which agree by reciprocity and the anti-involution. Therefore these generated polynomials are necessary and sufficient for associativity. Removing zero polynomials and normalizing signs of repeated equations changes no solution.

3.2 Safe pruning, including singular linear systems

The baseline solver starts with a finite domain for every variable. It evaluates an enclosing interval for each polynomial and rejects a branch only if that interval excludes zero. If an equation contains at most two unfixed variables, it tests every pair of values in their current domains and removes unsupported values. These operations preserve every solution.

Equations that have become linear are row-reduced over $\mathbb{F}_{257}$. An inconsistent modular system cannot have an integer solution. Consequences involving one or two variables restrict their finite domains. Rank deficiency by itself never rejects a branch: free variables remain in the exhaustive search. Since $M \leq 15 < 257$, a residue determines at most one permitted integer. The final leaf is always checked against every original integer polynomial.

Lexicographic comparison with the duality centralizer retains the least coordinate vector in each orbit. A comparison stops at its first undecided coordinate; one cannot infer a later lexicographic inequality past an earlier unknown entry. Any extra prefix inequalities are necessary conditions for the same least representative.

Theorem 3.1 (Conditional completeness of a terminated search). *Fix r , M and a duality type. If the finite-domain solver terminates normally and processes every branch of its search tree, it emits exactly one tensor from each based-isomorphism class satisfying that type and bound. If every duality type terminates normally, their union is the complete whole-rank census.*

Proof. Every admissible tensor has an orbit-coordinate vector in the initial box. Each pruning rule is a necessary condition, so the least representative of that tensor is never deleted. At each unpruned node the first unfixed variable is assigned every remaining value. Induction on the number of unfixed variables therefore visits each feasible least vector. Leaf tests prove soundness, and lexicographic minimality proves uniqueness. Finally, duality types are disjoint and exhaust all possibilities. $\square$

A parallel run partitions a fixed coordinate prefix into all bounded assignments. The resulting subtrees are disjoint and their union is the original tree. A run is marked complete only after every prefix finishes. A time limit is not a mathematical rejection rule: a timeout leaves the larger census uncertified.

4 Self-dual reconstruction from one multiplication matrix

Put $n = r - 2$. Choosing a nonunit element gives a symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & a & u^\top \\ 0 & u & Q \end{pmatrix}, \quad u \in \mathbb{Z}_{\geq 0}^n, \quad Q = Q^\top \in M_n(\mathbb{Z}_{\geq 0}).$$

The remaining parameters are a symmetric cubic tensor on the n tail indices. Relabelling allows u to be nondecreasing. If $b = u_1$ is the least mixed repeated-index coefficient, then every such coefficient is at least b ; in particular $Q_{ii} \geq b$. Ties in u allow the corresponding diagonal entries of Q to be sorted. These are necessary conditions on an orbit minimum, not extra axioms imposed on the ring.

4.1 The two contraction identities

Let R_i be the tail block of multiplication by the i th tail basis element. Comparing the lower blocks of its commutator with A gives

$$R_i u = (Q^2 + uu^\top - I - aQ)e_i, \tag{4}$$

$$R_i Q = QR_i - (Qe_i)u^\top + u(Qe_i)^\top. \tag{5}$$

Writing $s = \sum_j u_j$, each contraction lies in $[0, Ms]$ and is divisible by $\gcd(u_1, \dots, u_n)$, with zero required if $u = 0$. Mixed-coefficient bounds strengthen the diagonal lower bound to $b(s - u_i)$ and the off-diagonal lower bound to $b(u_i + u_j)$. Partial sums of squares in Q^2 are enclosed by letting each missing entry range over $[0, M]$. These necessary interval and divisibility conditions eliminate many first matrices before cubic completion.

4.2 A finite-field reconstruction with an exhaustive fallback

Define

$$V_k = Q^k u, \quad K = [V_0, \dots, V_{n-1}], \quad H_0 = Q^2 + uu^\top - I - aQ, \\ H_{k+1} = QH_k - (u^\top V_k)Q + uV_{k+1}^\top.$$

Induction using (5) proves $R_i V_k = H_k e_i$. Each H_k is affine in a . If K is invertible modulo a prime $p > M$, then

$$R_i = [H_0 e_i, \dots, H_{n-1} e_i] K^{-1} \pmod{p}.$$

Each reconstructed residue has at most one representative in $[0, M]$. Retaining only those representatives and then checking all original integer equations gives an exact reconstruction, not an approximate numerical method.

The implementation tries $p = 257$ and $p = 17$. A preliminary row system can reject values of a without reconstructing all tail entries. If either system has free coordinates, its compatibility equations are retained. With one or two free coordinates, all their permitted values are tested. With more free coordinates the general exhaustive completion is used. If neither prime gives an invertible K , the solver searches every surviving integer value of a and every compatible remaining coordinate. In particular, an integer matrix that happens to be singular modulo a chosen prime is never lost.

4.3 Folding a fixed invertible element

When $u = 0$, $a = 0$ and $Q = I$, the chosen element g satisfies $g^2 = 1$ and $gX_i = X_i$ for all tail elements. The tail product then has the form

$$X_i X_j = \delta_{ij}(1 + g) + \sum_k t_{ijk} X_k.$$

Its associativity equations are the rank- $(r - 1)$ symmetric equations with the unit contribution changed from one to two. This is an auxiliary weighted system, not a new fusion ring. Replace I by $2I$ in the contraction formulae, enumerate that smaller system, and unfold. Each unfolded tensor is minimized under the original relabellings and checked against the original equations.

The distinguished pair $\{1, g\}$ is intrinsic: another invertible element h fixed by g would yield $g = 1$ after multiplying $gh = h$ by h^{-1} . Thus folding introduces no hidden identifications. It is used only in this proved case; weighted systems are not recursively folded under an unproved extension of the argument.

5 What the independent verification establishes

The enumerator and verifier have separate mathematical tasks. Exhaustiveness uses the complete coordinate parameterization, the safe-pruning proof and the normal completion of every required branch. The verifier reads only the emitted *full tensors*; it neither reconstructs the parameterization nor trusts the generator's polynomial list.

For every tensor it checks nonnegative integral coefficients, both unit laws, uniqueness and involutivity of the dual, both reciprocity identities, and every integer associativity identity for all four indices. It then canonicalizes under *every* unit-fixing basis permutation. A matching hash is only a lookup aid: actual canonical tensors are compared. This detects both literal repetitions and different labellings of an isomorphic based ring.

Passing these checks proves soundness and uniqueness of the emitted data; by itself it would not prove that no ring was omitted. That is why normal completion logs and the mathematical coverage argument are retained separately. Comparison with the original ancillary file is an additional containment/regression test, not an input to enumeration and not a replacement for an exhaustiveness proof.

5.1 Provenance and reproducibility

The retained Vercleyen–Slingerland v4 ancillary file has SHA-256

`b587c3f683157369da7cc090f762bbff0e18dcf11acc3f5031d6bc0c643843f8`

Its 28,451 records contain 25,138 distinct classes. The 3,313 redundant records occur at rank five and exact multiplicities 9–12. Their original and corrected counts are respectively

$$(1463, 1794, 2283, 3049) \longrightarrow (863, 1082, 1383, 1948).$$

Every deletion has an explicit basis-permutation witness in the audit folder. This audit does not establish completeness of the source’s partial-search cells.

The primary manifest `results/census.json` records every released count, bound, filename and checksum. Compressed tensors use the original line-per-ring, nested-curly-brace format. An entry is $N[i][j][k] = N_{ij}^k$: the output index has already been dualized where the internal parameter convention requires it. The raw tensor needs no additional index conversion.

5.2 Computation budgets and release gates

The private repository provides two manually dispatched GitHub Actions campaigns. The original frontier has a 1,100-second shared driver deadline and a 20-minute job ceiling. The long frontier selects only ranks 5–8: each rank has one 4,200-second (70-minute) shared deadline and a 75-minute job ceiling, at most 300 runner-minutes in total. Compilation, all attempted multiplicities and duality types, export, compression and independent verification share the driver deadline. Enumeration receives 90% of remaining driver time; the balance is reserved for other phases. Threads follow the actual runner CPU allocation.

Only a complete whole-rank run with successful independent verification enters `best/`. A later incomplete attempt never replaces it. Candidates require explicit review and fresh independent verification before integration. Partial logs remain diagnostic evidence and never supply exhaustive census or OEIS terms. All workflows are manual; neither a push nor this manuscript launches a search.

The reviewed long campaign, GitHub run 35440299307 at source revision 931addc686e8 (full revision in the review record), completed rank five through multiplicity 19 with 34,133 classes, including 5,640 at exact multiplicity 19. The subsequent rank-five bound 20 and the attempted bounds 7, 4 and 2 at ranks six, seven and eight exhausted their search budgets. They yield no larger complete whole-rank counts. Ranks three and four were not recomputed. The accepted rank-five data underwent a fresh exhaustive tensor and based-isomorphism audit on GitHub before integration. The full earlier exact-multiplicity prefix agrees. See `docs/CAMPAIGN_REVIEW.md` and `verification/imports/rank5_through19/`.

6 Reading and reusing the data

The result files count based rings, not monoidal categories or realizations of a ring. One tensor can admit several inequivalent categorifications, or none. No number here is a classification of fusion categories. The rank-specific count files use exact multiplicity; cumulative totals belong only in the summary column. Proposed OEIS additions must be regenerated from a complete manifest, not copied from an interrupted log.

The retained code was generated with AI assistance and is accompanied by explicit mathematical coverage arguments and independent integer checks. These checks are not a formal proof in a proof assistant. The present text is an explanatory computational companion, not a claim of journal acceptance or external peer review.

7 Reproduction commands

From the repository root, a complete run can be requested as follows. A timeout reports failure to finish, not a smaller theorem.

```
python3 code/run_census.py --rank 6 --bound 6 \
  --seconds 4200 --threads "$(nproc)" --verify --out rerun_rank6
```

To verify the supplied release without recomputing it:

```
python3 scripts/check_release.py --full
```

For a shared-budget frontier search, use `scripts/extend_census.py`. For ranks five through eight it starts at the released bound plus one and restarts the unfinished bound without claiming checkpoint resumption. The low-level command above limits enumeration only; the frontier wrapper additionally bounds every phase. Rank three is instead capped at 1000.

References

- [1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015.
- [2] G. Vercleyen and J. Slingerland, *On Low Rank Fusion Rings*, Journal of Mathematical Physics 64 (2023), 091703; [arXiv:2205.15637v4](#).
- [3] V. Ostrik, *On formal codegrees of fusion categories*, Mathematical Research Letters 16 (2009), 895–901; [arXiv:0810.3242](#).
- [4] J. Dong and S. Palcoux, rank-four and rank-five census derivations, project manuscripts and computational supplements, September 2026. The relevant derivations are retained in the repository’s `methods/` directory.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A348305, A354471, A354472, A354473, A354475, A354476 and A354477. Proposed updates in this repository are not submitted edits.